

Sentimental Text Analysis Pipeline for different kind of Medias

From uploads to sentiment-ready tokens

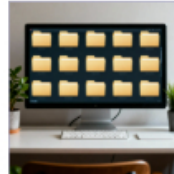
Secure upload, OCR/PyPDF/Whisper extraction, normalized cleaning, tokenization, and rule-aware sentiment scoring.

GAGAN K M



Automated Text Extraction and Sentiment Pipeline

Upload to sentiment insight with OCR, PyPDF, Whisper, and preprocessing



Input ingestion: Image, PDF, Audio

- Image or PDF or audio file uploaded
- Supports common formats for batch or single files

Text extraction methods

- OCR extracts text from images using OCR pipeline
- PyPDF extracts text from PDFs
- Whisper transcribes audio to text

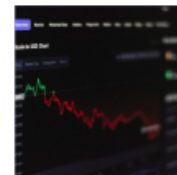


Preprocessing: cleaning and normalization

- Remove punctuation, numbers, and symbols
- Convert text to lowercase and strip whitespace

Tokenization and sentiment analysis

- Cleaned text tokenized for model input
- Sentiment analysis assigns polarity scores
- Stopwords detection lowers score and flags detail



OCR Pipeline for Images: Options, Preprocessing, Modes

Compare engines, optimize input, measure errors, and mitigate common failures

■ OCR Options

Tesseract, Google Vision API, AWS Textract, .

■ Known Failure Modes

Handwriting, complex layouts, watermarks, low resolution, multi-column misalignment.

■ Downstream Flow

Upload → OCR/PyPDF/Whisper → clean (lower, remove punctuation, numbers, symbols) → tokenize → sentiment analysis.

■ Sentiment Edge Case

If stopwords dominate, assign negative score and display detected terms.

PDF Text Extraction with PyPDF

Decision flow for reliable text extraction
from PDFs

01 Direct Extraction
Use PyPDF for selectable text with correct
encodings

02 Code Considerations
Handle rotated pages, multiple encodings,
preserve reading order

03 Common Issues
Ligatures, hyphenation, missing fonts, metadata
mismatches, images treated as text

04 Processed to next steps
Passed the extracted contents to Preprocessing





Audio Transcription with Whisper Model

Guidance on model choice, sampling.



01

Model variants

Whisper tiny/base/small/medium/large —
trade latency versus accuracy and language
coverage.

02

Conversion to Texts

The Audio is Transcribed into Texts

03

Passing to Next Pre Processing

passing the transcribed texts to further
preprocessing

Text Preprocessing: Lowercase, Strip, Remove Noise

Normalize case, clean characters, then tokenize for sentiment

01

Normalize Case

Lowercase all texts

02

Remove Punctuation, Numbers, Symbols

Apply regex or NLP libraries to remove noisy characters for sentiment models.

03

Tokenize the texts

Python: regex, NLTK, spaCy, clean-text – integrate snippets into pipeline.

04

Analyse Sentimentally and Score calculation

Validate the Sentiment Score



Tokenization : Whitespace versus Subword

How tokenizer choice affects sentiment model
input and outcomes

Whitespace Tokenizer

1. Simpler and faster with full-word vocabulary
2. Out of vocabulary words remain whole
3. Poor handling of rare or misspelled tokens

Subword Tokenizer (BPE / WordPiece)

1. Handles OOV via subword units for better generalization
2. Smaller vocabulary size, better coverage of morphology
3. Increases token count for long or compound words

Sentiment Analysis: Models, Scoring, Thresholds, Interpretability

Choose models and thresholds that balance accuracy, actionability, and explainability

use Transformers for analysis

Rule-based lexicons, ML classifiers, or transformer models for nuanced sentiment

Thresholding and Alerts scoring for stopwards

Automated actions at defined thresholds; flag scores for human review

Define Score and Detailing

provide an complete information of score, stopward and result.

Stopword-Based Negative Scoring Logic

Detect, weight, and flag
stopword-driven negative
sentiment

Detection Stopword Logic

Match tokens to curated
stopword list; count occurrences
and compute weighted penalty.



Weighting Rules

Higher weight for contextual
stopwords; scale penalty relative
to text length.



Triggering Negative Score

If penalty over threshold,
reduce sentiment by penalty and
annotate reasons for review.

